

RAS SEDR, Egypt

WMO: 624550

Lat: 29.5844N

Lon: 32.7175E

Elev: 16

StdP: 101.13

Time Zone: 2.00 (E02)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.7	9.8	-3.1	2.9	15.7	-0.8	3.5	16.1	9.7	16.3	8.0	17.1	1.7	0	0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.5	38.3	21.8	37.2	21.7	36.2	21.7	25.1	31.8	24.5	31.2	24.0	30.7	3.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.1	28.4	22.6	17.4	28.1	22.0	16.8	28.0	76.8	32.1	74.4	31.2	72.4	30.6	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.6	6.8	6.1	DB	6.0	41.3	1.9	1.4	4.6	42.3	3.5	43.2	2.4	44.0	1.0	45.0	
			WB	3.0	26.8	1.5	1.3	1.9	27.7	1.1	28.5	0.2	29.2	-0.9	30.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.3	14.9	16.3	18.9	22.2	25.9	29.0	30.7	30.8	28.4	25.0	20.7	16.7	
	DBStd	5.95	1.94	2.33	2.69	2.70	2.63	2.11	1.54	1.34	1.65	2.03	2.27	1.81	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	261	109	66	24	1	0	0	0	0	0	0	4	56	
	CDD10.0	4867	151	176	277	366	494	569	642	646	552	466	322	208	
	CDD18.3	2087	2	8	43	116	236	319	383	387	302	208	76	6	
	CDH23.3	24466	4	39	258	916	2524	4109	5368	5457	3653	1769	354	14	
	CDH26.7	12224	0	4	47	281	1146	2238	3078	3096	1734	554	47	0	
Wind	WSAvg	3.1	2.4	2.5	3.2	3.4	3.5	3.6	3.4	3.5	3.5	3.1	2.5	2.3	
Precipitation	PrecAvg	13	3	2	3	0	0	0	0	0	0	1	2	1	
	PrecMax	29	12	8	24	3	7	1	0	0	0	6	9	8	
	PrecMin	4	0	0	0	0	0	0	0	0	0	0	0	0	
	PrecStd	7	3	2	5	1	1	0	0	0	0	1	2	2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.7	27.0	30.2	33.7	37.9	39.5	39.9	39.1	37.2	33.8	30.0	25.0	
		MCWB	15.6	16.2	17.1	17.3	19.8	20.8	22.1	22.5	21.3	21.7	18.9	16.9	
	2%	DB	21.6	24.4	27.8	31.4	35.3	37.6	38.3	37.8	35.4	31.9	28.0	23.0	
		MCWB	14.5	15.0	16.7	17.2	19.4	20.9	22.0	22.6	21.9	21.6	18.9	15.4	
	5%	DB	20.2	22.5	26.1	29.7	33.6	36.3	37.0	36.8	34.2	30.7	26.5	21.9	
		MCWB	13.4	14.2	15.9	16.9	19.1	20.9	21.9	22.4	22.0	21.2	18.5	14.9	
	10%	DB	19.2	21.0	24.4	28.1	32.0	34.9	35.9	35.6	33.1	29.6	25.3	21.0	
		MCWB	12.9	13.8	15.3	16.6	18.8	20.8	21.9	22.4	22.0	20.9	17.8	14.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.4	18.4	20.2	20.3	21.9	24.2	25.6	26.1	25.8	24.7	22.0	18.5
			MCDB	21.9	23.1	25.4	28.1	31.6	32.1	33.6	32.7	30.9	29.5	26.6	22.9
2%		WB	15.4	16.2	18.0	19.0	21.0	23.3	24.7	25.2	24.4	23.5	20.8	16.8	
		MCDB	20.0	22.2	25.5	27.3	30.9	31.0	32.0	31.8	30.7	29.2	25.6	21.6	
5%		WB	14.4	15.2	17.0	18.2	20.4	22.7	24.1	24.7	23.8	22.6	19.7	15.8	
		MCDB	19.3	21.1	24.1	26.5	30.0	30.7	31.0	31.3	30.1	28.2	24.9	20.8	
10%		WB	13.4	14.3	16.0	17.4	19.8	22.1	23.6	24.2	23.3	21.9	18.7	14.9	
		MCDB	18.5	20.1	22.9	25.8	29.1	30.8	30.8	30.9	29.7	27.7	24.1	20.0	
Mean Daily Temperature Range	5% DB	MDBR	8.3	8.7	9.9	10.9	11.7	11.9	11.4	10.5	9.8	8.9	8.8	8.5	
		MCDBR	9.5	10.7	11.9	12.7	13.3	13.2	12.3	11.5	10.9	9.9	9.6	9.5	
		MCWBR	6.2	6.1	5.9	5.3	4.7	4.3	4.0	3.5	3.8	4.3	5.4	6.0	
	5% WB	MCDBR	9.1	9.6	10.7	11.3	11.9	12.0	11.5	10.7	10.1	8.9	8.6	9.0	
		MCWBR	6.4	6.0	5.9	5.3	4.6	4.3	4.0	3.6	3.9	4.3	5.2	6.2	
Clear-Sky Solar Irradiance	taub	0.384	0.421	0.472	0.525	0.537	0.452	0.448	0.446	0.455	0.463	0.397	0.377		
	taud	2.165	2.045	1.894	1.786	1.784	2.032	2.067	2.078	2.044	2.012	2.178	2.207		
	Ebn at Noon	831	832	813	785	775	841	842	841	819	778	810	819		
	Edh at Noon	126	154	192	221	223	173	166	163	163	158	123	115		
All-Sky Solar Radiation	RadAvg	3.73	4.67	6.07	7.18	7.92	8.44	8.21	7.61	6.61	5.27	4.13	3.50		
	RadStd	0.16	0.20	0.23	0.15	0.21	0.12	0.13	0.12	0.16	0.15	0.15	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+0.54	N/A	-0.67	+0.58	N/A	N/A	N/A	-50	+212	+171

Nomenclature: See separate page